Database Engine

Functional Components and Transactions

Database Engine

A database system is divided into modules, each handling a set of responsibilities. The main functional components are:

- **Storage Manager**
- **Query Processor**
- **Transaction Management**

1.6.1 Storage Manager

Definition

The storage manager provides the interface between the low-level data stored in the database and the application programs or queries submitted to the system.

Note

Tasks of the storage manager:

- Interaction with OS file manager
- Efficient storage, retrieval, and updating of data
- Enforcing authorization and integrity constraints
- Transaction management for consistency

Components of Storage Manager

- **Authorization & Integrity Manager:** Checks user authority and integrity constraints.
- **Transaction Manager:** Ensures consistency despite system failures.
- **File Manager:** Allocates space and maintains disk structures.
- **Buffer Manager:** Fetches data from disk to main memory and manages caching.

Data Structures Implemented by Storage Manager

- **Data files:** Store the database relations.
- **Data dictionary:** Stores metadata about database structure (schemas).
- **Indices:** Provide fast access to data values.

1.6.2 Query Processor

Definition

The query processor interprets, optimizes, and executes database queries.

Components of Query Processor

- **DDL Interpreter:** Interprets DDL statements and updates the data dictionary.
- **DML Compiler:** Converts DML statements into evaluation plans.
- **Query Evaluation Engine:** Executes low-level instructions from the DML compiler.

Query Processing Steps

1. Parsing and Translation
2. Optimization (lowest-cost evaluation plan)
3. Evaluation (execution of instructions)

1.6.3 Transaction Management

Definition

A **transaction** is a collection of operations that performs a single logical function in a database application. **Example:** deposit, withdrawal, transfer between accounts.

ACID Properties of Transactions

- **Atomicity:** All operations in a transaction are completed or none.
- **Consistency:** Database remains consistent before and after transaction.
- **Isolation:** Concurrent transactions do not interfere.
- **Durability:** Committed changes are permanent, even after failures.

Transaction Manager Components

- **Concurrency Control Manager:** Ensures consistency during concurrent transactions.
- **Recovery Manager:** Maintains consistent database state during system or transaction failures.

Example

Example of Concurrency Control: Two users accessing the same bank account cannot withdraw more than allowed or corrupt the system.